

IDS Code

CJD

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Assignment brief • Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:

1. Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries.
2. By 2020, substantially reduce the proportion of youth not in employment, education or training.

Find another CSV file to add to analysis

Loading libraries:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

We must set the wd and import the 3 CSV files as data frames. (note eet means education, employment, and training)

```
setwd("/Users/charliedicker/Downloads/data_sets")

continents <- read.csv("continents-according-to-our-world-in-data.csv")
gdp_per_capita_df <- read.csv("gdp-per-capita-worldbank.csv")
youth_not_in_eet <- read.csv("youth-not-in-education-employment-training.csv")

# We now create a vector of all LDC countries
ldc_countries <- c(
  "Afghanistan", "Angola", "Bangladesh", "Benin", "Bhutan", "Burkina Faso",
  "Burundi", "Cambodia", "Central African Republic", "Chad", "Comoros",
```

```

"Democratic Republic of the Congo", "Djibouti", "Eritrea", "Ethiopia",
"Gambia", "Guinea", "Guinea-Bissau", "Haiti", "Kiribati",
"Lao People's Democratic Republic", "Lesotho", "Liberia", "Madagascar",
"Malawi", "Mali", "Mauritania", "Mozambique", "Myanmar", "Nepal",
"Niger", "Rwanda", "Sao Tome and Principe", "Senegal", "Sierra Leone",
"Solomon Islands", "Somalia", "South Sudan", "Sudan", "Timor-Leste",
"Togo", "Tuvalu", "Uganda", "United Republic of Tanzania",
"Vanuatu", "Yemen", "Zambia"
)

# I downloaded population data off world bank, to calculate gdp
population_df <- read.csv("population.csv")

```

We now clean the data, getting rid of any irrelevant data

```

# The year is irrelevant in continents, as the only thing this df needs to be used for is assigning a c
continents$Year <- NULL
continents$Entity <- NULL

# We now assign the continent to each country in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  left_join(continents, by = "Code")

# Same thing for youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  left_join(continents, by = "Code")

# The continents table is now redundant
rm(continents)

# We clean population data, dropping all columns that aren't year and country
population_df$Series.Name <- NULL
population_df$Series.Code <- NULL
population_df$Country.Code <- NULL

```

Still cleaning (renaming columns to nicer names)

```

# Rename a column in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  rename(Gdp_per_capita = GDP.per.capita..PPP..constant.2017.international...)

# Rename a column in youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  rename(Perc_youth_not_in_eet = Share.of.youth.not.in.education..employment.or.training..total...of.y

population_df <- population_df %>%
  rename(
    Year_2015 = X2015..YR2015.,
    Year_2016 = X2016..YR2016.,
    Year_2017 = X2017..YR2017.,
    Year_2018 = X2018..YR2018.,

```

```

Year_2019 = X2019..YR2019.,
Year_2020 = X2020..YR2020.,
Year_2021 = X2021..YR2021.,
Entity = Country.Name
)

```

Dropping values before the year 2014, and if population data is above 2021, and if continent is NA.

```

# Drop all values before 2014, as these UN goals were set in 2015
gdp_per_capita_df <- gdp_per_capita_df %>%
  filter(Year >= 2014 & !is.na(Continent))

youth_not_in_eet <- youth_not_in_eet %>%
  filter(Year >= 2014 & !is.na(Continent))

```

Here we make population into long format, so that we can join the data frames

```

population_df <- population_df %>%
  pivot_longer(
    cols = starts_with("Year_"),
    names_to = "Year",
    values_to = "Population"
  ) %>%
  mutate(
    Year = as.integer(sub("Year_", "", Year)) # remove "Year_" and make numeric
  )

```

Calculating Economic Growth and change in eet of each country

```

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Growth = ifelse(lag(Entity) != Entity, NA, (Gdp_per_capita - lag(Gdp_per_capita)) / lag(Gdp_per_capita)))

youth_not_in_eet <- youth_not_in_eet %>%
  mutate(Perc_change = ifelse(lag(Entity) != Entity, NA, (Perc_youth_not_in_eet - lag(Perc_youth_not_in_eet)) / lag(Perc_youth_not_in_eet)))

```

This chunk is now to add a flag to the gdp per capita dataframe, labelling if a country is an LDC or not

```

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Ldc = Entity %in% ldc_countries)

```

Merging the data frames to include population in both

```

gdp_per_capita_df <- merge(gdp_per_capita_df, population_df, by = c("Entity", "Year"))
youth_not_in_eet <- merge(youth_not_in_eet, population_df, by = c("Entity", "Year"))

```

Now we convert to numeric and calculate GDP per capita of each country each year

```

gdp_per_capita_df$Population <- as.numeric(gdp_per_capita_df$Population)

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Gdp = Population * Gdp_per_capita)

```

Finally, export the CSV files of the 2 data frames

```
write.csv(gdp_per_capita_df, "/Users/charliedicker/Documents/Uni/Year 1/R basics (data science)/Coursework/2025/11/25/gdp_per_capita.csv")
write.csv(youth_not_in_eet, "/Users/charliedicker/Documents/Uni/Year 1/R basics (data science)/Coursework/2025/11/25/youth_not_in_eet.csv")
```

REF: 25/11/2025 asked ChatGPT “what would the definition of”least developed countries” mean in terms of gdp per capita (PPP)”

26/11/2025 asked ChatGPT “give me a list of all countries the UN considers LDCs between 2015 and 2021, with each country in quotation marks seperated by commas”